Using Elastic File System With Instances.

1. What is Amazon EFS

Amazon Elastic File System (EFS) is a **fully managed Network File System (NFS)** provided by **Amazon Web Services** that allows **multiple EC2 instances to access the same storage simultaneously**.

Instead of attaching storage to just one instance (like EBS), EFS acts like **shared network storage**.

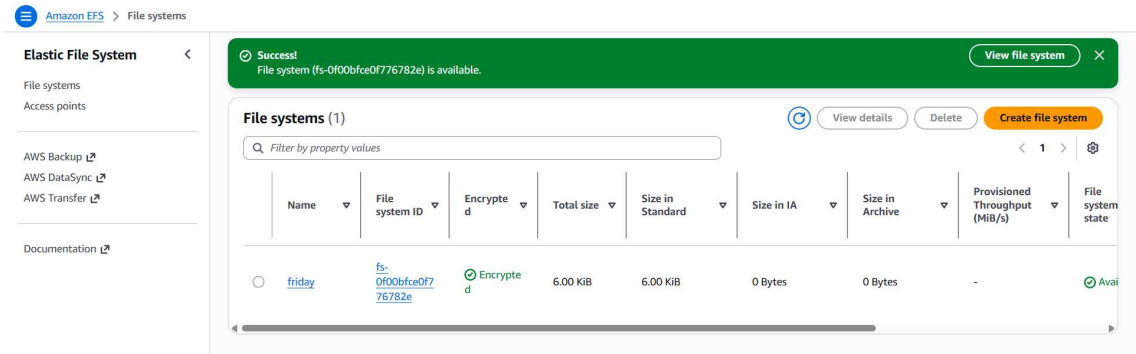
Key Idea

- One file system
- Many EC2 instances
- All read/write the same files

This is commonly used in **web servers, container workloads, and shared application data**.

I have created 3 instances and an EFS to connect with.

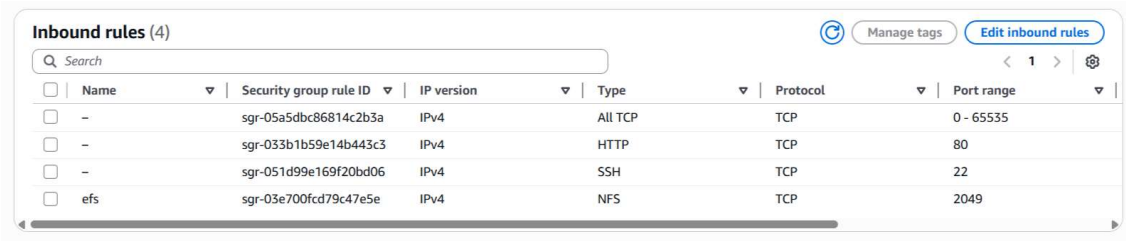
EFS



To create 3 Instances

Then we have to make sure that the both for EFS and EC2's Security Group must have to allow the NFS port.

For Instance.



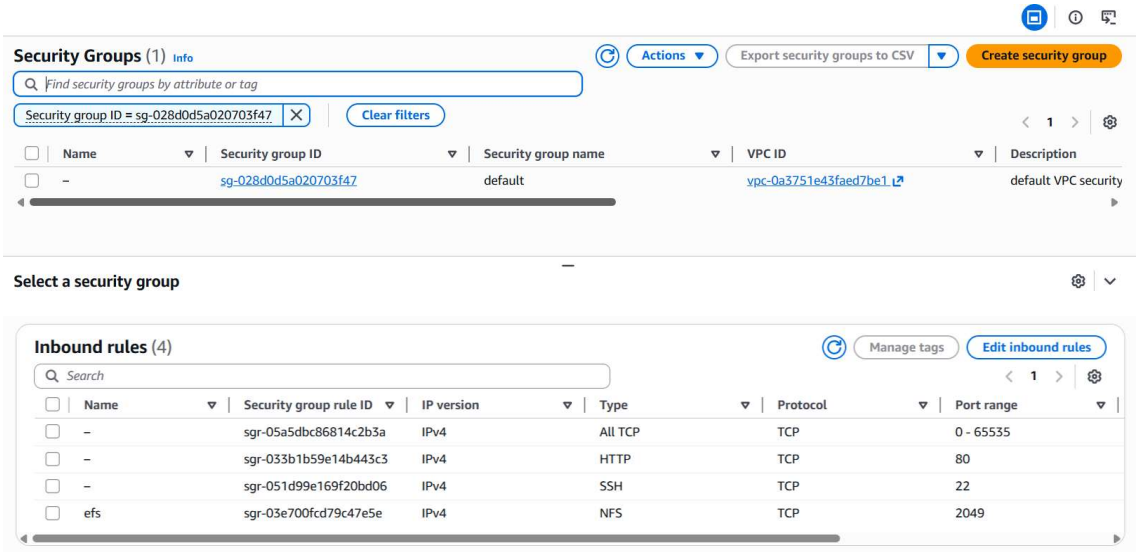
Inbound rules (4)

Manage tags | Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range
–	sgr-05a5dbc86814c2b3a	IPv4	All TCP	TCP	0 - 65535
–	sgr-033b1b59e14b443c3	IPv4	HTTP	TCP	80
–	sgr-051d99e169f20bd06	IPv4	SSH	TCP	22
efs	sgr-03e700fcd79c47e5e	IPv4	NFS	TCP	2049

For EFS.



Security Groups (1) Info

Actions | Export security groups to CSV | Create security group

Find security groups by attribute or tag

Security group ID = sg-028d0d5a020703f47 | Clear filters

Name	Security group ID	Security group name	VPC ID	Description
–	sg-028d0d5a020703f47	default	vpc-0a3751e43faed7be1	default VPC security

Select a security group

Inbound rules (4)

Manage tags | Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range
–	sgr-05a5dbc86814c2b3a	IPv4	All TCP	TCP	0 - 65535
–	sgr-033b1b59e14b443c3	IPv4	HTTP	TCP	80
–	sgr-051d99e169f20bd06	IPv4	SSH	TCP	22
efs	sgr-03e700fcd79c47e5e	IPv4	NFS	TCP	2049

Then we have to copy the command given in the Attach EFS option.

We have to change it with our created file name.

```
[ec2-user@ip-172-31-45-160 ~]$ sudo su
[root@ip-172-31-45-160 ec2-user]# mkdir sunday
[root@ip-172-31-45-160 ec2-user]# sudo mount -t nfs4 -o r
.amazonaws.com:/ sunday
[root@ip-172-31-45-160 ec2-user]# cd sunday/
[root@ip-172-31-45-160 sunday]# ls
mayday
[root@ip-172-31-45-160 sunday]#
```

```
        _/m/'
[ec2-user@ip-172-31-45-160 ~]$ sudo su
[root@ip-172-31-45-160 ec2-user]# mkdir sunday
[root@ip-172-31-45-160 ec2-user]# sudo mount -t nfs4 -o r
.amazonaws.com:/ sunday
[root@ip-172-31-45-160 ec2-user]# cd sunday/
[root@ip-172-31-45-160 sunday]# ls
mayday
[root@ip-172-31-45-160 sunday]# touch rural
[root@ip-172-31-45-160 sunday]# ls
mayday  rural
[root@ip-172-31-45-160 sunday]#
```

Then I have do the same thing in the another instance.

And I have got the file that have been created in the main instance to another instance.

```
[ec2-user@ip-172-31-34-156 ~]$ sudo su
[root@ip-172-31-34-156 ec2-user]# mkdir monday
[root@ip-172-31-34-156 ec2-user]# ls
monday
[root@ip-172-31-34-156 ec2-user]# sudo mount -t nfs4 -o nfsve
.amazonaws.com:/ monday
[root@ip-172-31-34-156 ec2-user]# ls
monday
[root@ip-172-31-34-156 ec2-user]# cd monday/
[root@ip-172-31-34-156 monday]# ls
mayday  rural
[root@ip-172-31-34-156 monday]#
```

The files are replicated on the other connected instance

```
[ec2-user@ip-172-31-34-156 ~]$ sudo su
[root@ip-172-31-34-156 ec2-user]# mkdir monday
[root@ip-172-31-34-156 ec2-user]# ls
monday
[root@ip-172-31-34-156 ec2-user]# sudo mount -t nfs4 -o r
.amazonaws.com:/ monday
[root@ip-172-31-34-156 ec2-user]# ls
monday
[root@ip-172-31-34-156 ec2-user]# cd monday/
[root@ip-172-31-34-156 monday]# ls
mayday  rural
[root@ip-172-31-34-156 monday]# touch city
[root@ip-172-31-34-156 monday]# ls
city  mayday  rural
[root@ip-172-31-34-156 monday]#
```

```
[ec2-user@ip-172-31-45-160 ~]$ sudo su
[root@ip-172-31-45-160 ec2-user]# mkdir sunday
[root@ip-172-31-45-160 ec2-user]# sudo mount -t nfs4 -
.amazonaws.com:/ sunday
[root@ip-172-31-45-160 ec2-user]# cd sunday/
[root@ip-172-31-45-160 sunday]# ls
mayday
[root@ip-172-31-45-160 sunday]# touch rural
[root@ip-172-31-45-160 sunday]# ls
mayday  rural
[root@ip-172-31-45-160 sunday]# ls
city  mayday  rural
[root@ip-172-31-45-160 sunday]#
```

```
Last login: Thu Mar 12 06:11:08 2026 from 13.233.177.5
[ec2-user@ip-172-31-38-72 ~]$ sudo su
[root@ip-172-31-38-72 ec2-user]# mkdir jarvis
[root@ip-172-31-38-72 ec2-user]# sudo mount -t nfs4 -o nfsvers=
amazonaws.com:/ jarvis
[root@ip-172-31-38-72 ec2-user]# ls
jarvis
[root@ip-172-31-38-72 ec2-user]# cd jarvis/
[root@ip-172-31-38-72 jarvis]# ls
[root@ip-172-31-38-72 jarvis]# touch mayday
[root@ip-172-31-38-72 jarvis]# ls
mayday
[root@ip-172-31-38-72 jarvis]# ls
city  mayday  rural
[root@ip-172-31-38-72 jarvis]#
```